Safety and Efficacy of Nafcillin for Empiric Therapy of Late-Onset Sepsis in the NICU

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abstract

BACKGROUND AND OBJECTIVE: In 2014 at Nationwide Children's Hospital, the Neonatal Antimicrobial Stewardship Program recommended nafcillin over vancomycin for empirical therapy of possible late-onset sepsis (LOS) in infants without a history of methicillin-resistant *Staphylococcus aureus* colonization or infection. We report our experience with this guideline and assess its safety.

METHODS: We retrospectively reviewed all infants who received nafcillin or vancomycin for empirical treatment of possible LOS at 3 NICUs before (January 2013–May 2014) and after (January 2017–March 2019) implementation of a vancomycin reduction guideline. Safety measures included duration of blood culture positivity, recurrence of infection with the same previously identified pathogen in the 14 days after discontinuation of antibiotic therapy, and mortality.

RESULTS: Among 366 infants who received a first antibiotic course for possible LOS, 84% (95 of 113) and 25% (62 of 253) received empirical therapy with vancomycin before and after the guideline implementation, respectively, representing a 70% reduction. Nafcillin use increased by 368%. Duration of blood culture positivity did not differ before and after the guidance. In 2 infants, antibiotic therapy was restarted within 14 days of discontinuation of the initial therapy for recurrence of the same infection; both had received empirical vancomycin. Overall in-hospital mortality was 10%, and there was no difference before (9%) and after (10%) implementation of the vancomycin reduction guidance (odds ratio, 0.97).

CONCLUSIONS: Nafcillin can be a safe alternative to vancomycin for empirical therapy of LOS among NICU infants who do not have a history of methicillin-resistant *S aureus* infection or colonization.

Full article can be found online at www.pediatrics.org/cgi/doi/10.1542/peds.2021-052360

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This work was presented in part at IDWeek; October 2–6, 2019; Washington, DC.

DOI: https://doi.org/10.1542/peds.2021-052360

WHAT'S KNOWN ON THIS SUBJECT: Vancomycin, in combination with an aminoglycoside, has been recommended for empirical treatment of late-onset sepsis in NICUs, as coagulase-negative staphylococci remain the most frequent bloodstream pathogens. Concern exists for development of vancomycin resistance from its overuse.

WHAT THIS STUDY ADDS: The Neonatal Antimicrobial Stewardship Program recommendation that nafcillin be used for late-onset sepsis reduced overall vancomycin utilization by 67% without changing the duration of culture positivity, recurrence of same infection within 14 days after discontinuation of antibiotic therapy, and mortality.

To cite: Magers J, Prusakov P, Speaks S, et al. Safety and Efficacy of Nafcillin for Empiric Therapy of Late-Onset Sepsis in the NICU. *Pediatrics.* 2022;149(5):e2021052360

With the survival of very low birth weight infants in the 1980s, coagulase-negative staphylococci (CoNS) emerged as the predominant pathogens causing late-onset sepsis (LOS) among preterm infants in the NICU.[1,2] Because most CoNS isolates are resistant to β-lactam antibiotics, including the penicillinase-resistant penicillin agents, vancomycin in combination with an aminoglycoside became the preferred agent for suspected LOS in the NICU.[3] As vancomycin use increased in the United States and globally,[4] resistance to vancomycin emerged with methicillin-resistant *Staphylococcus aureus* (MRSA) and *Enterococcus faecium* colonization and infection.[5–7] Outbreaks of vancomycin-resistant enterococci along with reports of reduced vancomycin susceptibility among methicillin-resistant CoNS in NICUs added to the emergent need to reduce vancomycin usage.[8–15]

Beginning in 2014 at Nationwide Children's Hospital in Columbus, Ohio, the Neonatal Antimicrobial Stewardship Program (NEO-ASP) team recommended the use of nafcillin instead of vancomycin, in combination with gentamicin, for empirical therapy of possible LOS in infants without a history of MRSA colonization or infection (Fig 1). Our objective in the current study is to report our experience with this guideline and assess the safety of the practice. Our hypothesis was that vancomycin use can be reduced safely in the NICU.

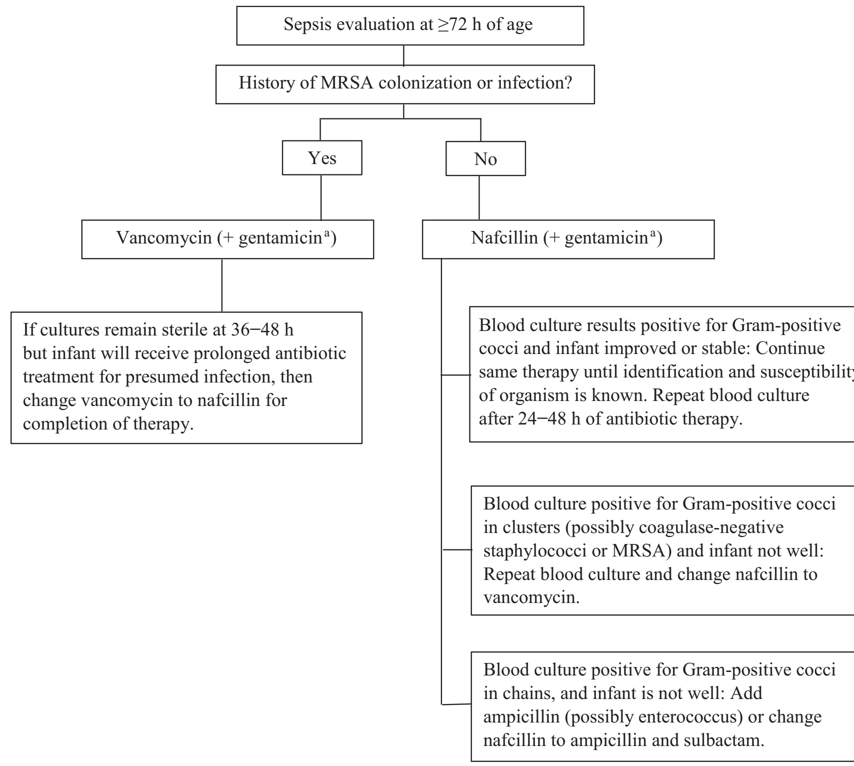

FIGURE 1
Vancomycin reduction guideline for empirical therapy of possible LOS in infants in the NICU. [a] If Gram-negative meningitis, add third- or fourth-generation cephalosporin agent.

METHODS

We performed a retrospective review of all infants who received nafcillin, vancomycin, or both for empirical treatment of possible LOS at 3 Nationwide Children's Hospital NICUs in Columbus, Ohio, from January 1, 2013, to May 31, 2014 (preintervention), and January 1, 2017, to March 31, 2019 (postintervention), before and after the implementation of the vancomycin reduction guideline, respectively. The period of June 1, 2014, to December 31, 2016, was excluded from the analysis to allow for the introduction and implementation of the vancomycin reduction guidance at the 9 NICUs in Columbus. The 3 largest NICUs that also have the highest acuity and the same electronic health record (EPIC) were selected for analysis of the vancomycin reduction guidance. The NICU at Nationwide Children's Hospital is a level 4 outborn referral NICU, whereas the other 2 affiliated NICUs at Riverside Methodist Hospital and Grant Medical Center are level 3 and located in delivery hospitals. These 3 NICUs include 190 beds with ~1900 admissions annually. The study was approved by the Nationwide Children's Hospital Institutional Review Board with a waiver of informed consent.

Per the guidance (Fig 1), infants without a history of MRSA colonization or infection and who were evaluated for possible LOS at >72 hours of age received empirical treatment with nafcillin rather than vancomycin, irrespective of clinical or laboratory parameters such as the presence of a central venous catheter, hypotension, or abnormal acute phase reactants. Infants who were known to be colonized with MRSA at the time of the sepsis evaluation received empirical vancomycin treatment. Subsequent antibiotic therapy was based on results of cultures obtained as part of the sepsis evaluation. Infants with necrotizing enterocolitis or spontaneous intestinal perforation were excluded from analysis because their initial antibiotic therapy per protocol was ampicillin and sulbactam in combination with gentamicin.

Infants who received at least 1 dose of nafcillin or vancomycin after 72 hours of age in the 3 NICUs were identified by query of the electronic health record. Pertinent demographic, clinical, laboratory, and diagnostic data were obtained, including gestational age, birth weight, sex, chronologic age at the time of initiation of antibiotic therapy, MRSA colonization or infection history, and presence of a central venous catheter at the time of the sepsis evaluation. Evaluation for suspected sepsis and components of the evaluation were at the discretion of the attending neonatologist, as were decisions regarding duration of specific antimicrobial agents. Symptoms of sepsis, such as temperature instability, irritability or lethargy, apnea, bradycardia, or desaturation; respiratory support and oxygen requirement; and hemodynamic instability requiring intravenous fluid bolus(es) or vasopressor therapy were noted. Laboratory tests at the time of the sepsis evaluation, if performed, included C-reactive protein, white blood cell count and differential, and platelet count, as well as dates and results of blood or cerebrospinal fluid cultures and whether these were obtained before or after antibiotic administration.

The first dose of either nafcillin or vancomycin provided to infants in the NICU was used to define the initiation of the sepsis evaluation, irrespective of whether any infant sample was obtained for bacterial culture. Sepsis evaluations were classified and analyzed as being the first evaluation for possible sepsis (first antibiotic course) and by total sepsis evaluations (all antibiotic courses). The dates that antibiotics were initiated and discontinued and the length of antibiotic therapy, defined as any calendar day that the infant received vancomycin or nafcillin, were recorded. Safety measures included duration of culture positivity as determined by days to culture sterilization, recurrence of infection with the same previously identified pathogen in the 14 days after discontinuation of antibiotic therapy, and mortality. In addition, infection-related in-hospital mortality was assessed for all sepsis evaluations by review of the attending physician's documentation in the electronic health record or the autopsy report, if available.

A proven infection was defined by isolation of a pathogen from blood or cerebrospinal fluid obtained at >72 hours of age along with clinical symptomatology. A possible infection was defined as clinical symptomatology with an accompanying blood culture that was sterile or yielded a possible contaminant, but the infant was treated >48 hours by the attending physician. If antibiotic therapy was provided for ≤48 hours, the clinical scenario was classified as a rule-out sepsis course.[16]

Statistical Analysis

Medians with interquartile ranges (IQRs) were used to summarize continuous variables, and counts with percentages were used for categorical data. Log-binomial regression was used to calculate incidence rate ratios (IRRs) to compare the change in antibiotic use from the pre- to postintervention periods for the first antibiotic course and all antibiotic courses. Interrupted time-series analysis for nafcillin and vancomycin was used to estimate the rate of antibiotic initiation by month in the pre- and postintervention periods. IRRs also were calculated to compare infants who had a central venous catheter and MRSA colonization in the pre- and postintervention periods. Univariable logistic regression models were used to estimate the odds ratios (ORs) of mortality separately for the pre- and postintervention periods for vancomycin and nafcillin. A third logistic regression model with intervention period, antibiotic, and the interaction was used to calculate estimates of the OR of mortality by antibiotic for the pre and post periods as well as estimates of the probability of mortality within each group (eg, preintervention vancomycin, preintervention nafcillin, postintervention vancomycin, and postintervention nafcillin). Cluster-robust SEs[17] were used in these models to account for the clustering within infants. Additional covariates were not included in these models because the study aim was to estimate the association of antibiotic with mortality regardless of other covariate values.[18] Furthermore, overfitting and convergence would be a concern because of the very low rate of mortality.[19] Duration of positive culture results was compared using the Kruskal-Wallis rank sum test. Microsoft Excel or R version 4.1.1 (R Core Team 2021; R Foundation for Statistical Computing) were used for data management, descriptive statistics, and figures. Analyses were performed using SPSS for Windows version 26.0 (IBM Corporation, Armonk, NY), R (R Core Team 2021), and Stata release 16 (StataCorp LLC, College Station, TX).

RESULTS

During the 2 study periods, 366 infants (Table 1) underwent 516 sepsis evaluations (Table 2). Empirical therapy with nafcillin or vancomycin was initiated in 209 (57%) and 157 (43%) infants for 277 (54%) and 239 (46%) sepsis evaluations, respectively. The majority of the infants were male (n = 209, 57%) and preterm (median gestational age, 28 weeks; IQR, 25–34 weeks) with a median birth weight of 1002 g (IQR,

TABLE 1 Characteristics of the 366 Infants Comprising the Study Population Before and After Implementation of the Vancomycin Reduction Guidance in the NICU

	Empirical Antibiotic Therapy			
	Preintervention (January 1, 2013, to May 31, 2014)		Postintervention (January 1, 2017, to March 31, 2019)	
	Nafcillin	Vancomycin	Nafcillin	Vancomycin
First sepsis evaluation				
Infants	18 (16)	95 (84)	191 (75)[a]	62 (25)[b]
Gestational age, wk	28 (25–32)	27 (25–34)	29[c] (26–36)	26[c] (24–32)
Birth weight, g	878 (601–1946)	1001 (703–2305)	1139[c] (765–2414)	820[c] (626–1240)
Male	9 (50)	54 (57)	109 (57)	37 (60)
Age, d	48 (9–80)	14 (7–44)	19 (11–39)	14 (6–41)
MRSA colonization/infection	0	10 (11)	1 (1)	10 (16)
CVC	6 (33)	57 (60)	98 (51)	39 (63)
CVC duration, d	4 (3–7)	6 (3–8)	7 (4–12)	5 (3–12)
Subsequent sepsis evaluations, No.	10	47	58	35
Age, d	96 (48–182)	47 (22–92)	52 (29–91)	71 (41–117)
MRSA	0	6 (13)	3 (5)	14 (40)
CVC	4 (40)	26 (55)	17 (29)	19 (54)
CVC duration (days)	48 (17–80)	13 (7–25)	11 (6–18)	16 (10–36)
Total antibiotic courses	28 (16)	142 (84)	249 (72)[d]	97 (28)[e]
Overall mortality	4 (22)	6 (6)	17 (9)	9 (15)
Infection-related mortality[f]	0	0	1 (0.5)[g]	3 (5)[h]

Data are presented as *n* (%) or median (IQR) unless otherwise indicated. CVC, central venous catheter.
[a] A 368% increase in empirical nafcillin use.
[b] A 70% reduction in empirical vancomycin use for initial sepsis evaluation.
[c] Statistical differences in pairwise analysis.
[d] A 350% increase in empirical nafcillin use.
[e] A 67% decrease in empirical vancomycin use for all sepsis evaluations.
[f] Comparison of infection-related mortality between empirical nafcillin and vancomycin for the postintervention period.
[g] Chest tube wound culture positive for MRSA.
[h] One blood culture positive for *E faecalis*, 1 blood culture positive for *K pneumoniae*, and 1 disseminated adenovirus infection.

720–2305 g); 10% (36 of 366) of infants died during the NICU hospitalization.

Evaluation of the first antibiotic course for possible LOS during the preintervention period of 15 months revealed that 84% (95 of 113) of infants received empirical therapy with vancomycin and 16% (18 of 113) received nafcillin. In the postintervention period, 25% (62 of 253) of infants received vancomycin, whereas 75% (191 of 253) received nafcillin. The use of nafcillin compared with vancomycin for first antibiotic course was higher in the postintervention period than the preintervention period (IRR, 4.74; 95% CI, 3.01% to 7.96%; Table 3).

The total antibiotic courses initiated with nafcillin in the preintervention period was 16% (28 of 170) and with vancomycin, 84% (142 of 170), whereas in the postintervention period, initiation with nafcillin was 72% (249 of 346) and with vancomycin, 28% (97 of 346). Evaluation of all antibiotic courses revealed that the use of nafcillin compared with vancomycin was higher in the postintervention period than in the preintervention period (IRR, 4.37; 95% CI, 3.01% to 6.60%; Table 3). The difference in the percentage of infants receiving nafcillin in the postintervention period was 59.5% (95% CI, 50.6% to 68.1%) compared with the preintervention period. Overall, the combined total incidence of vancomycin and nafcillin use decreased from 3.85 starts per 1000 patient-days to 2.76 starts per 1000 patient-days. The estimated rate of initiation for nafcillin increased by 1.2 per 1000 patient-days (95% CI, 0.18 to 2.18 patient-days; $P = .02$). The estimated rate of initiation for vancomycin decreased by 2.4 per 1000 patient-days (95% CI, −4.96% to 0.14%, $P = .06$). The change in slope comparing the pre- and postintervention periods for nafcillin was 0.02 (95% CI, −0.09% to 0.05%) and for vancomycin, 0.08 (95% CI, −0.09% to 0.25%; Fig 2).

The number of infants who had a central venous catheter at the time of the first sepsis evaluation was not different between the preintervention (56%, 63 of 113) and postintervention periods (54%, 137 of 253; IRR, 0.97; 95% CI, 0.72% to 1.32%; Table 3). During the postintervention period, the number of infants with central venous catheters did not differ between those who received empirical nafcillin (51%, 98 of 191) or vancomycin (63%, 39 of 62). On the other hand, on subsequent sepsis evaluations, colonization with

TABLE 2 Definitive Antibiotic Therapy and Bacteriologic Results Among 366 Infants Who Had 516 Sepsis Evaluations Before and After Implementation of the Vancomycin Reduction Guidance in the NICU

	Definitive Antibiotic Therapy			
	Preintervention (January 1, 2013, to May 31, 2014)		Postintervention (January 1, 2017, to March 31, 2019)	
	Nafcillin	Vancomycin	Nafcillin	Vancomycin
Infants	18 (16)	95 (84)	191 (75)	62 (25)
Sepsis evaluations	28 (16)	142 (84)	249 (72)	97 (28)
Treatment courses	13 (30)	31 (70)	44 (61)	28 (39)
Treatment indication				
Sepsis culture positive	2 (15)	6 (19)	14 (32)	15 (54)
Sepsis culture negative	3 (23)	12 (39)	9 (20)	3 (11)
Meningitis[a]	0	1 (3)	4 (9)	0
Pneumonia	3 (23)	9 (29)	12 (27)	4 (14)
Urinary tract infection	4 (31)	0	3 (7)	3 (11)
Other	1 (8)	3 (10)	2 (5)	3 (11)
Pathogen (blood)				
Gram-positive	1 (4)	7 (5)	17 (7)	16 (16)
CoNS	1 (100)	3 (43)	7 (41)	12 (75)
MSSA	0	4 (57)	8 (47)	1 (6)
MRSA	0	0	1 (6)	2 (13)
E faecalis	0	0	0	1 (6)
S pneumoniae	0	0	1 (6)	0
Gram-negative	2 (7)	0	0	0
P aeruginosa	1 (50)	0	0	0
K pneumoniae	1 (50)	0	0	0
Duration of positive culture, d, median (IQR)[b]	1 (1–2)	4 (2–6)	2 (1–3)	1 (1–3)
Length of therapy, d, median (IQR)[b]	7 (6–8)	8 (7–10)	8 (7–9)	9 (8–12)
Antibiotic changed during therapy	4 (31)	8 (26)	11 (25)	5 (18)
Recurrence of same infection within 14 d ofdiscontinuation of antibiotic therapy	0	1 (3)	0	1 (4)

Data are presented as *n* (%) unless otherwise indicated.
[a] All bacterial cultures of cerebrospinal fluid were sterile; diagnosis was based on abnormal indices.
[b] Refers to Gram-positive isolates only. Duration of positive culture among all 4 groups was not statistically different (Kruskal-Wallis rank sum test $P = .12$).

MRSA was higher during the postintervention period (18% [17 of 93] vs 11% [6 of 57]; Table 1), and as per the guideline, more of these infants received vancomycin therapy.

Of the total 516 sepsis evaluations, 400 (78%) resulted in discontinuation of antibiotic therapy within 48 hours of initiation. The remaining 116 (22%) sepsis evaluations were representative of definitive antibiotic therapy and consisted of continuation or change in the empirical antibiotic coverage, depending on culture results or clinical status (Table 2). During both time periods, the majority of pathogens detected in blood were Gram-positive bacteria, mostly CoNS. Thirteen Gram-negative bacteria were identified during the study periods, but patients were not continued on nafcillin or vancomycin except for 2. Of the 40 CoNS blood isolates, 23 (58%) were assessed as causing proven infection, whereas 91% (30 of 33) of isolates that had undergone susceptibility testing were resistant to methicillin. Of the 16 bloodstream infections due to *S aureus*, 13 (81%) were susceptible to nafcillin.

During the course of the sepsis episode, 15 infants who received nafcillin empirically were treated subsequently with vancomycin. Seven (47%) infants had a proven infection with CoNS, whereas 5 (33%) received prolonged therapy without an identified pathogen. Two (13%) other infants had nafcillin changed to vancomycin because of a lack of clinical improvement, even though the blood culture detected *Pseudomonas aeruginosa* in 1 infant and *Klebsiella pneumoniae* in the other. One additional patient had bacteremia with *Streptococcus pneumoniae* and received a treatment course of vancomycin. Thirteen infants who received vancomycin empirically were later changed to nafcillin during the same sepsis episode, with 5 (38%) having methicillin-susceptible *S aureus* (MSSA) bacteremia, 2 having methicillin-susceptible CoNS bacteremia, and 6 treated without an identified pathogen detected in blood culture.

Only 2 infants had antibiotic therapy restarted within 14 days of discontinuation of the initial therapy for recurrence of the same infection (Table 2). Both infants had received

TABLE 3 Model Estimates for Nafcillin Use, Central Venous Catheter Incidence, MRSA Colonization, and Mortality by Intervention Period or Use of Vancomycin or Nafcillin

	Estimate (95% CI)	*P*
Count of nafcillin use		
First course only (IRR)		
Preintervention use of nafcillin	Reference	
Postintervention use of nafcillin	4.74 (3.01 to 7.96)	<.001
All courses (IRR)[a]		
Preintervention use of nafcillin	Reference	
Postintervention use of nafcillin	4.37 (3.01 to 6.60)	<.001
Central venous catheter (IRR)		
Preintervention	Reference	
Postintervention	0.97 (0.72 to 1.32)	.08
MRSA colonization (IRR)		
Preintervention	Reference	
Postintervention	0.49 (0.21 to 1.17)	.10
Mortality (OR)[a]		
Preintervention[b]	Reference	
Postintervention[b]	0.97 (0.4 to 2.34)	.95
Vancomycin[c]	Reference	
Nafcillin[c]	0.99 (0.5 to 1.93)	.97
Preintervention period[d]		
Vancomycin	Reference	
Nafcillin	2.12 (0.64 to 6.98)	.22
Postintervention period[d]		
Vancomycin	Reference	
Nafcillin	0.75 (0.33 to 1.70)	.49
Probability of mortality[d]		
Preintervention/nafcillin	0.09 (0.02 to 0.16)	—
Preintervention/vancomycin	0.18 (0.01 to 0.34)	—
Postintervention/nafcillin	0.13 (0.05 to 0.20)	—
Postintervention/vancomycin	0.10 (0.05 to 0.14)	—

—, not applicable.

[a]Cluster-robust SEs were used to account for repeated antibiotic courses within individual infants.

[b] Estimates from a univariable logistic regression model with intervention as the covariate.

[c] Estimates from a univariable logistic regression model with antibiotic as the covariate.

[d] Estimates from a multivariable logistic regression model with intervention period and antibiotic type and their interaction.

empirical vancomycin therapy, with 1 in the preintervention period and the other in the postintervention period. No infant who received nafcillin empirically or for definitive treatment had antibiotic therapy restarted for recurrence of the same infection in the ensuing 14 days after discontinuation of the initial antibiotic therapy.

In-hospital mortality during the 2 study periods was 10% (36 of 366) and did not differ before (9%, 10 of 113) and after (10%, 26 of 253) implementation of the vancomycin reduction guideline (OR, 0.97; 95% CI, 0.4% to 2.34%; Tables 1 and 3). There also was no difference in mortality among infants who received empirical nafcillin or vancomycin (OR, 0.99; 95% CI, 0.5% to 1.93%; Table 3). A multivariable logistic regression with intervention period and antibiotic type (data not shown) had estimated ORs similar to the unadjusted ORs. No infant in the preintervention period and 4 (1.6%) in the postintervention period died as a result of an infection-related cause (Table 1). From the multivariable logistic regression model with intervention period and antibiotic and their interaction, the OR of mortality in the preintervention period comparing vancomycin with nafcillin was 2.12 (95% CI, 0.64% to 6.98%), and the OR of mortality in the postintervention period comparing vancomycin with nafcillin was 0.74 (95% CI, 0.32% to 1.70%; Table 3). The estimated probability of mortality for each combination of antibiotic and time period is shown in Table 3. Of the deaths, 1 infant who received empirical nafcillin had a sterile blood culture result, but the chest tube wound culture yielded MRSA, and antibiotic therapy was changed to vancomycin (Table 1). Results of the surveillance cultures that detected MRSA were known after the sepsis evaluation was performed. Among the 3 infants who had received empirical vancomycin treatment, 1 with gestational age of 22 weeks had a vancomycin-susceptible *E faecalis* bloodstream infection, 1 had a *K pneumoniae* bloodstream infection, and 1 had disseminated adenoviral sepsis. No infant died secondary to infection with CoNS.

DISCUSSION

In this retrospective review of 3 NICUs of Nationwide Children's Hospital, we found that implementation of a vancomycin reduction guidance spearheaded by a multidisciplinary NEO-ASP committee resulted in the safe and sustained reduction in its use, despite CoNS being the most frequent pathogen. Empirical use of nafcillin rather than vancomycin was not associated with reinitiation of antibiotic therapy within 14 days of its discontinuation and did not result in higher overall or infection-related in-hospital mortality. In addition, it did not affect duration of bacterial blood culture positivity.

Vancomycin, in combination with an aminoglycoside, has been frequently recommended for empirical treatment of LOS in NICUs worldwide, as CoNS remain the most frequent bloodstream pathogens.[20] Concern for development of vancomycin resistance among Gram-positive

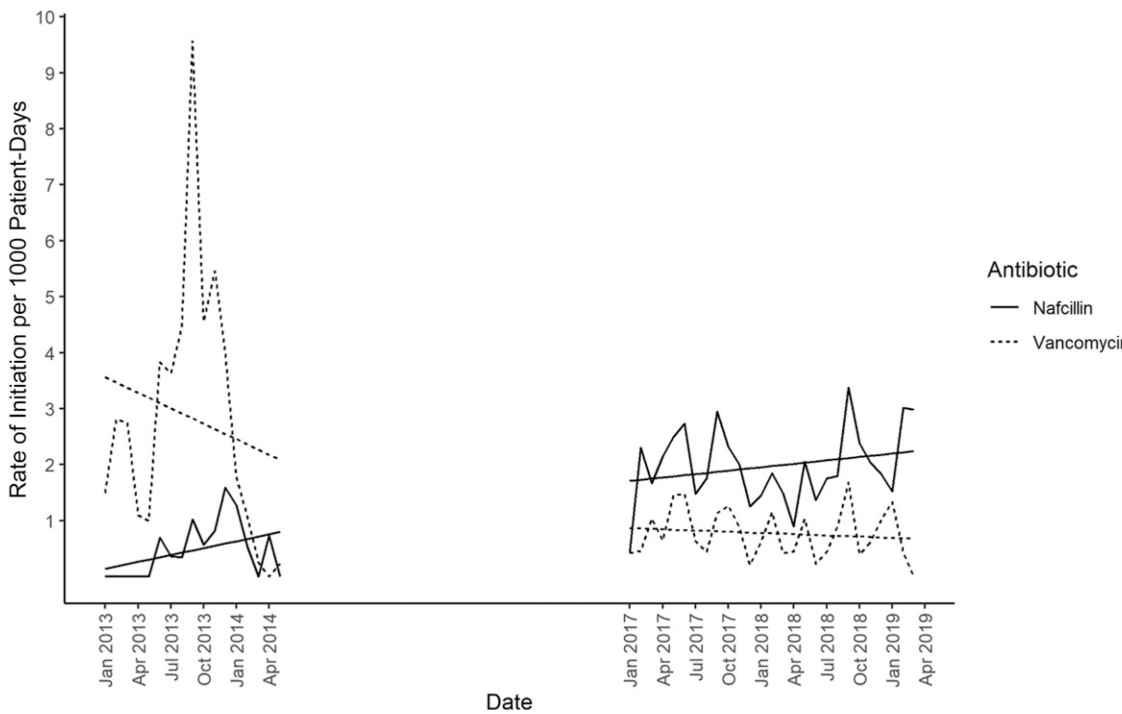

FIGURE 2
Interrupted time-series analysis of initiation of vancomycin and nafcillin for possible LOS in 3 NICUs of Nationwide Children's Hospital before (January 1, 2013, to May 31, 2014) and after (January 1, 2017, to March 31, 2019) implementation of a vancomycin reduction guidance. The estimated rate of initiation for nafcillin increased by 1.2 per 1000 patient-days (95% CI, 0.18% to 2.18%, $P = .02$). The estimated rate of initiation for vancomycin decreased by 2.4 per 1000 patient-days (95% CI, −4.96% to 0.14%, $P = .06$). The change in slope comparing the pre- and postintervention period for nafcillin was −0.02 (95% CI, −0.09% to 0.05%; $P = .54$) and for vancomycin, 0.08 (95% CI, −0.09% to 0.25%; $P = .33$).

organisms and minimizing potential toxicity has resulted in the call for judicious use of vancomycin.[21–27] Efforts to reduce the empirical use of vancomycin have been hampered by the fact that CoNS are almost universally resistant to other agents, including the β-lactamase–resistant penicillins such as nafcillin.[28] Studies in North and South America and Europe have revealed that vancomycin reduction can be done safely and without changes in mortality, duration of bacteremia, or complications attributable to LOS.[21,23–25,29–35] Unlike in the current study, however, Ericson et al[36] found that the median duration of bacteremia with CoNS was 1 day longer in infants who received delayed vancomycin therapy. However, empirical vancomycin therapy was not associated with improved survival. Bacteremia with CoNS is rarely if ever fulminant,[37–39] and the timely identification of these organisms in neonatal blood cultures allows for the institution of appropriate vancomycin therapy within 24 hours of the sepsis evaluation.

Most antibiotic usage in the NICU is for empirical therapy.[16] Therefore, our strategy to reduce overall vancomycin usage targeted initiation and not just discontinuation of initial therapy when cultures did not yield a pathogen only susceptible to vancomycin. In contrast to infections due to CoNS, those due to MRSA carry a high mortality rate.[40,41] We therefore perform surveillance for MRSA in our NICUs and use empirical vancomycin therapy for infants with MRSA colonization or infection. As others have reported and seen in this study, MSSA has become more prevalent than MRSA in NICUs in the United States, and empirical nafcillin would be a more suitable agent than vancomycin for optimal treatment of MSSA while minimizing potential nephrotoxicity and ototoxicity.[42,43] Additionally, past exposure to vancomycin has been associated with Gram-negative bacilli bloodstream infections in very low birth weight infants and children.[44–47]

Limitations of this study include its retrospective nature and changes in infection prevention practices during the 7-year study period that may have affected the safety of the vancomycin reduction guidance. These changes include the institution of the NEO-ASP committee, which monitored all antibiotic use in the NICU, as well as the implementation of central line–associated bloodstream infection bundles.[48] The NICU policy is performance of only 1 blood culture at the time of sepsis evaluation, and therefore, detection of CoNS in blood cultures of some infants may have been from skin contamination rather than a true infection. In this study, 42% of CoNS blood isolates were regarded as contaminants by the attending physician or NEO-ASP committee, and antibiotic therapy was discontinued at <48 hours. In addition, only pathogens detected in blood or cerebrospinal fluid were analyzed, and although this procedure may have underestimated the impact of other infections involving the urinary tract system or bone and joints, all nafcillin and vancomycin use was assessed regardless of the site of infection. The use of other antistaphylococcal antibiotics for empirical sepsis treatment was not assessed because it was infrequent and against the recommended guidance. Such information, however, would not have affected the overall decrease in vancomycin use. Use of central venous catheters and MRSA colonization that may have affected vancomycin initiation were not different between the pre- and postintervention periods, although the small sample size may have affected the results. Finally, morbidity from delayed treatment of CoNS bacteremia, such as bronchopulmonary dysplasia, retinopathy of prematurity, and neurodevelopmental outcomes, was not investigated. Nevertheless, both overall in-hospital and infection-related mortality did not differ before and after implementation of the vancomycin reduction guideline.

CONCLUSIONS

We have shown that nafcillin is a safe and effective alternative to vancomycin for empirical therapy of LOS among high-risk infants in the NICU who do not have a history of MRSA infection or colonization. Prudent and judicious vancomycin use must remain a global public health imperative in our NICUs.[49]

ACKNOWLEDGMENTS

The authors thank the members of the NEO-ASP Committee (Randy Miller, MD; Richard Moraille, MD; Anthony Thiele, MD; Natalie O. White, DO; Alexandra K. Medoro, MD; Joshua Watson, MD; Gregory Ryshen; Melinda Albertson, NNP; Caitlyn Schwirian, PharmD; Nathan Johnson-Roddenberry, NNP; and Maria Jebbia, MD) at Nationwide Children's Hospital for their support of the vancomycin reduction guideline.

ABBREVIATIONS

CI: confidence interval
CoNS: coagulase-negative staphylococci
IQR: interquartile range
IRR: incidence rate ratio
LOS: late-onset sepsis
MRSA: methicillin-resistant *Staphylococcus aureus*
MSSA: methicillin-susceptible *Staphylococcus aureus*
NEO-ASP: Neonatal Antimicrobial Stewardship Program
OR: odds ratio

PEDIATRICS (ISSN Numbers: Print, 0031-4005; Online, 1098-4275).

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PEDIATRICS (ISSN Numbers: Print, 0031-4005; Online, 1098-4275).

FUNDING: No external funding.

CONFLICT OF INTEREST DISCLOSURES: The authors have indicated they have no conflicts of interest relevant to this article to disclose.

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